

Jinwoo Cho

jimy12181218@gmail.com | linkedin.com/in/heyjinwoo | github.com/superjin1218 | Open to work

A data-driven, CS-backed researcher across finance, quant, data and machine learning — and blockchain. I design data collection–classification–analysis pipelines end to end and turn complex financial and market phenomena into conclusions anyone can act on.
Open to finance, data and research roles.

EDUCATION

Inha University

B.S. in Computer Engineering (in progress)

- GPA 3.96 / 4.5 · Certified Investment Manager (KOFIA, 2025)
- Focus: finance & quant (CGE econometrics), data pipelines, machine learning, market & on-chain research

Incheon, Korea

Mar 2023 – Present

Finance AI Lab, Inha University

Undergraduate Researcher · sites.google.com/view/inha-ai-f-lab

Incheon, Korea

Feb 2025 – Jul 2025

EXPERIENCE

BLUENODE — Inha University Blockchain Society

Research Team Lead & Onboarding Team Lead

- Lead the research & onboarding teams; own research direction and output quality

Incheon, Korea

Jan 2026 – Present

StockTalk Inha — Inha stock open-chat

Founder & Operator

- Grew a market-discussion & research-sharing community to 670+ members
- Run the 'جو식 이야기' stock blog (2022–) — equity/macro/on-chain analysis, 30,000+ cumulative views

Incheon, Korea

2024 – Present

I.T.T — Incheon Inter-University Stock Club

Team 3 Lead

Incheon, Korea

Mar 2025 – Dec 2025

Incheon Bugwang High School

Blockchain Teaching Assistant

Incheon, Korea

Apr 2026 – Present

Republic of Korea Air Force

Vehicle Operator · Discharged as Sergeant

Korea

2021 – 2023

HONORS & AWARDS

2025 K-SURE Big-Data Analytics Competition — Grand Prize (1st) · 2025 23rd Korea Economics Paper Competition — Excellence (K-content CGE) · 2025 Korea Digital Contents Society Summer Conf. — Silver (sigmoid stock prediction) · 2025 WorldQuant Challenge — Stage 2 · 2025 Advantech Korea AIoT — Finalist · 2024 DB GAPS Investment Competition — Finals · 2024 Growthon Ideathon — Grand Prize · 2024 Agthon — Excellence · 2023 CFA Institute Research Challenge — Completed (Kolmar Korea vs Cosmax) · 2023 Sangmyung-Inha AWS — Top 6 · 2026 Arbitrum Mini Hackathon — 1st (STARK zk Coprocessor)

SKILLS & ADDITIONAL

Finance · Quant	Asset management, market research, 13F flow analysis, parametric insurance & options, strategy backtesting (win-rate/return), CGE econometrics
Data · Research	Data pipelines (SEC EDGAR crawler), big data, Monte-Carlo simulation, clustering (HDBSCAN), data visualization
AI · ML	Random Forest, DANN (domain-adversarial nets), forecasting, AI agents/harness, extensive LLM & MCP experience
Blockchain · On-chain	Ethereum, on-chain data analysis, stablecoins, ZK/STARK, Stylus (Rust/WASM), Solidity
Languages · Additional	Python, C++ · Certified Investment Manager (KOFIA, 2025) · ROK Air Force, Sergeant, service completed (2021–2023)

SELECTED PROJECTS & RESEARCH

details on p. 2

- K-SURE Big-Data Analytics — Grand Prize (1st)** — Won 1st place (Grand Prize) for a trade/credit big-data analytics project.
drive.google.com/file/d/18HV8nxDmOaBk0ad7qhuwEX9JAc8WWKNr
- 13F Hedge-Fund Flow Pipeline** — Crawled/parsed SEC EDGAR 13F for 6,023 funds into 70+ CSVs by sector & mcap.
hanglestocks.tistory.com/136
- Jeju Typhoon Risk Index — Parametric Insurance + Option** — ML-predicted typhoon loss with an objective parametric insurance + weather option (Samsung Insurance Conf.).
github.com/superjin1218/Samsung_Insurance_Conference
- K-content Export — CGE Analysis (Excellence)** — Static CGE on 2020 SAM quantifying the macro impact of an export shock.
github.com/superjin1218/Hankyung_Economics_Paper_Contest
- WorldQuant Field-Similarity Explorer** — Built/deployed a web app HDBSCAN-clustering 1,008 alpha fields into 76 groups (Stage 2).
iqc-site.vercel.app
- AI Gas Classification — Robust to Sensor Drift** — Physics feature-eng + DANN for drift-robust gas classification (Advantech finalist).
github.com/superjin1218/Advantech_Conference
- x402 Payment-Risk Insurance + LLM Attack Agent (ACE)** — Actuarial design adversarially measuring & pricing agent-payment risk; ERC-8004 NFT seal.
i402.vercel.app
- Korean On-chain Repo System** — On-chain intraday-DvP repo design; Monte-Carlo shows ~₩3.03T/yr benefit.
drive.google.com/file/d/1xg-MrQOMHZtQsNxESA9s5sbZdXKPlxOs
- starkverifier — Arbitrum Hackathon (1st)** — On-chain gas benchmark of Poseidon/Merkle verification, Stylus vs Solidity — 1st

01 K-SURE Big-Data Analytics

AWARD · 2025 (1ST)

A trade/credit big-data analytics project that won 1st place (Grand Prize) at the K-SURE competition.

- **FOCUS** Defining a meaningful analytics question on trade/credit data and extracting actionable insight.
- **METHOD** Designed a collect-clean-model-visualize pipeline and structured the conclusions end to end.

drive.google.com/file/d/18HV8nxDmOaBk0ad7qhuwEX9JAc8WWKNNr

02 13F Hedge-Fund Flow Pipeline

DATA · PIPELINE

A data pipeline that crawls & parses SEC EDGAR 13F-HR to quantify quarterly hedge-fund flows at scale.

- **FOCUS** Sector / market-cap / top-20 / options fund flows across 6,023 common funds.
- **METHOD** A SEC-API crawler/parser producing 70+ CSVs; quarter-over-quarter diffs surface net buys/sells.

hanglestocks.tistory.com/136

03 Jeju Typhoon Risk Index — Parametric Insurance + Option

INSURANCE · QUANT

A parametric insurance plus a weather option that pays on an objective typhoon index (Samsung Insurance Conference).

- **FOCUS** Quantified risk from 1970–2023 losses and 12 meteorological variables; computed loss/combined ratios.
- **METHOD** Random-Forest prediction + binomial-tree option valuation for hedging. $R^2=0.54$, 24.2% loss ratio.

github.com/superjin1218/Samsung_Insurance_Conference

04 K-content Export — CGE Analysis

AWARD · ECONOMETRICS

A CGE analysis of the macro ripple effects of K-content exports — Excellence Award, Korea Economics Paper Competition.

- **FOCUS** How content exports propagate to GDP, employment and up/downstream industries.
- **METHOD** Built a 2020 SAM, ran a static CGE: \$100M export → \$2.9B GDP, ~2,800 jobs.

github.com/superjin1218/Hankyung_Economics_Paper_Contest

05 WorldQuant Field-Similarity Explorer

QUANT · WEB APP

Self-built & deployed a web app that clusters and visualizes the similarity of 1,008 alpha fields (WorldQuant Stage 2).

- **FOCUS** Alpha-field structure and redundancy via pnl/text-weighted similarity.
- **METHOD** HDBSCAN (0.85/0.15) → 76 clusters; graph / map / heatmap / groups views.

iqc-site.vercel.app

06 AI Gas Classification — Robust to Sensor Drift

ML · FINALIST

Keeps gas classification robust as sensors age (drift) via time-invariant learning — Advantech Korea finalist.

- **FOCUS** Generalization under sensor aging and physically meaningful feature design.
- **METHOD** Physics-based feature-eng (8/sensor, 128 total) + DANN (gradient reversal). UCI 10-batch, 6 gases.

github.com/superjin1218/Advantech_Conference

07 x402 Payment-Risk Insurance + LLM Attack Agent (ACE)

BLOCKCHAIN · INSURANCE · AI

An actuarial design pricing x402 agent/machine-payment risk, with an LLM attack agent that stress-tests it.

- **FOCUS** Failure/abuse scenarios, coverage, triggers and settlement; leveraging ERC-8004.
- **METHOD** 8 LHAA attack modules adversarially measure the risk rate; exposure rating prices it.

i402.vercel.app

08 Korean On-chain Repo System

BLOCKCHAIN · FINANCE · QUANT

Designed an on-chain intraday-DvP version of Korea's short-term funding (repo) market and validated its economic benefit.

- **FOCUS** Quantifying settlement risk, haircut and intraday-liquidity benefit.
- **METHOD** A 10,000-run Monte-Carlo simulation — ~₩3.03T/yr benefit, -91.7% settlement risk.

drive.google.com/file/d/1xg-MrQOMHZtQsNxSA9s5sbZdXKP1xOs

09 starkverifier — Arbitrum Mini Hackathon (1st)

BLOCKCHAIN · AWARD (1ST)

Implemented ZK-friendly ops (Poseidon/Merkle) in Stylus (Rust/WASM) vs Solidity and benchmarked on-chain gas — 1st place.

- **FOCUS** Cost efficiency and feasibility of running ZK-friendly ops on-chain.
- **METHOD** Built both behind one ABI and measured/compared on-chain gas under identical inputs.

github.com/superjin1218/starkverifier
